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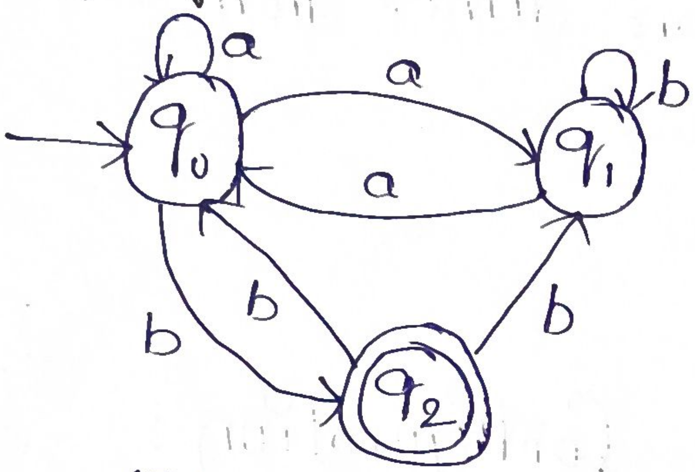
CSE-AI 192372204

1. Consider the NFA given below which accepts some language L over the alphabet $\Sigma = \{a, b\}$. Construct DFA that represents the same language L.

Sol



NFA

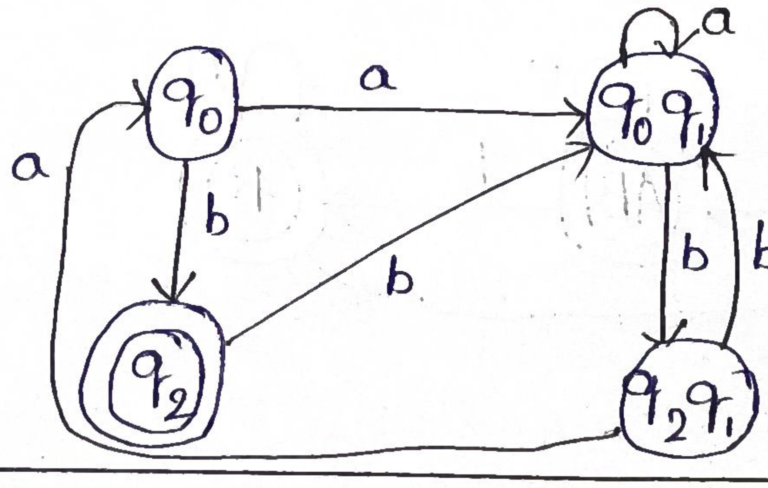
Configuration :

1. $\delta(q_0, a) = q_0 q_1$
2. $\delta(q_0 q_1, a) = q_0 q_1$
3. $\delta(q_0, b) = q_2$
4. $\delta(q_0 q_1, b) = q_2 q_1$
5. $\delta(q_2, a) = \emptyset$
6. $\delta(q_2, b) = q_0 q_1$
7. $\delta(q_2 q_1, a) = q_0$
8. $\delta(q_2 q_1, b) = q_0 q_1$

Conversion table :

state	a	b
q_0	$q_0 q_1$	q_2
$q_0 q_1$	$q_0 q_1$	$q_2 q_1$
q_2	$\emptyset$	$q_2 q_1$
$q_2 q_1$	q_0	$q_0 q_1$

DFA :

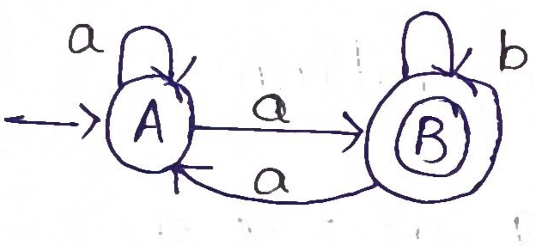


(2) Let L be the language of words of lenght atleast 2 whose last two letter are difference from each other. for example abaab is in L but not bb, Construct on NFA for L that draw the DFA

Sol. L = {last two letter are dfrnce from each other}

for example

L = abaab



NFA

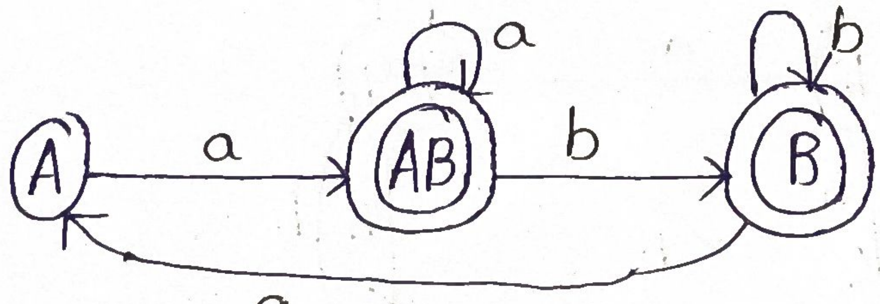
Configuration :

1. $\delta(A, a) = AB$
2. $\delta(A, b) = \emptyset$
3. $\delta(AB, a) = AB$
4. $\delta(AB, b) = B$
5. $\delta(B, a) = A$
6. $\delta(B, b) = B$

Conversion table :

State	a	b
A	AB	$\emptyset$
AB	AB	B
B	A	B

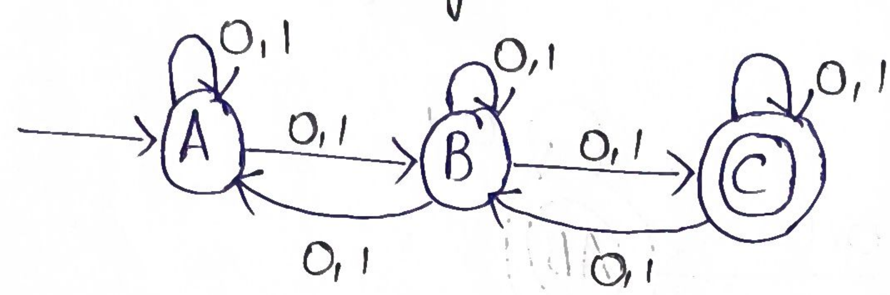
DFA



3. Design a NFA that takes a binary string as input and accept the string only if the last two digits are the same

ii) Write the formal defination of the NFA. Also show the string 01011 by the NFA

Sol



NFA

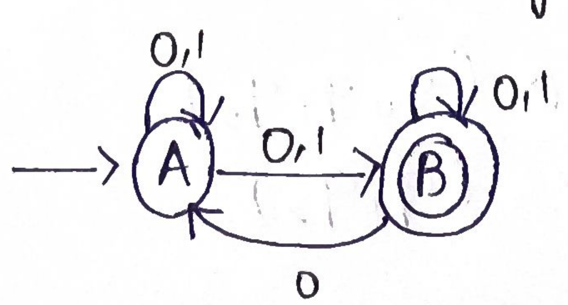
Take that given example:

01011

=> By using the above automata, we can accept the binary string.

4. Consider the NFA given below which accepts the set of all strings having 001 as a substring over the alphabet $\Sigma = \{0,1\}$ Construct a DFA that represents same language.

Sol



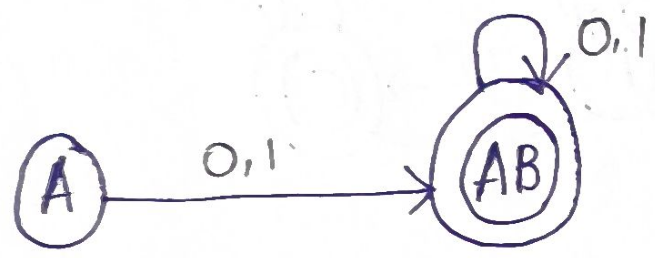
The above automata, accepts the set all substrings having 001.

Conversion table

state	0	1
A	AB	AB
AB	AB	AB

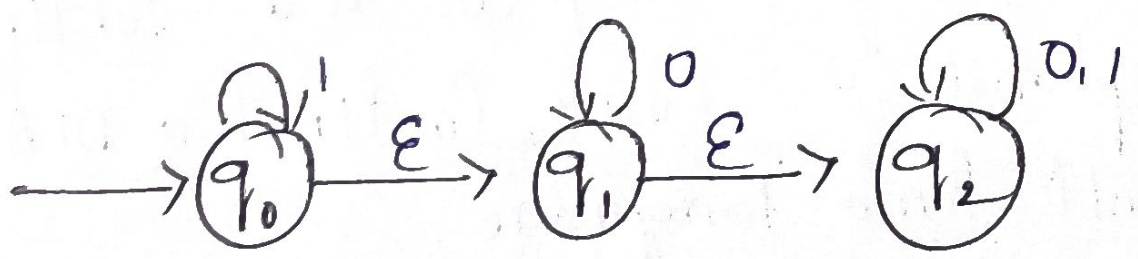
Configuration

1. $\delta(A,0) = AB$
2. $\delta(A,1) = AB$
3. $\delta(AB,1) = AB$
4. $\delta(AB,0) = AB$



DFA

(5) A NFA with ε-transitions is given below. find ε-closure for each state & design a equivalent DFA which accepts the same language. Draw its transition table & mark the initial & final states.



ε-Closure (A) = ABC $(q_0 q_1 q_2)$

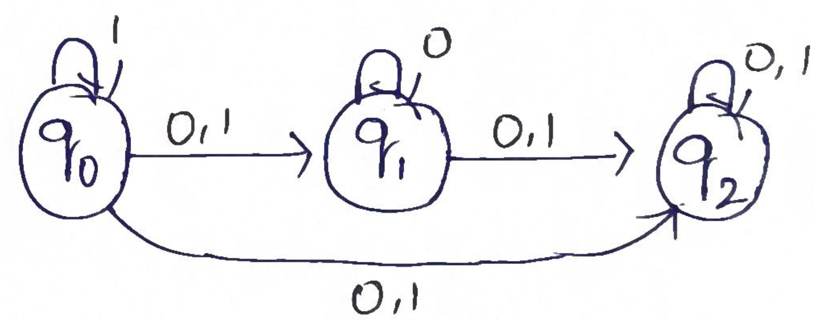
ε-Closure (B) = BC $(q_1 q_2)$

ε-Closure (q_2) = q_2

ε-Closure	State	0	1
$q_0 q_1 q_2$	q_0	$q_1 q_2$	$q_0 q_1 q_2$
$q_1 q_2$	q_1	$q_1 q_2$	q_2
q_2	q_2	q_2	q_2

<u>Configuration</u> :

1. $\delta(q_0, 0) = q_1 q_2$
2. $\delta(q_0\ 1) = q_0 q_1 q_2$
3. $\delta(q_1, 0) = q_1 q_2$
4. $\delta(q_1, 1) = q_2$
5. $\delta(q_2, 0) = q_2$
6. $\delta(q_2, 1) = q_2$



NFA

DFA